

5-Mark Questions & Answers -- Data Structures Module 2: Stacks & Queues

Q1. Write the complete algorithm and C functions for Infix to Postfix conversion using a Stack.

Detailed operator precedence algorithm handling ^ (highest, right-associative), *, /, +, -, and parentheses with complete C code.

Q2. Explain Circular Queue operations (Enqueue, Dequeue) with array implementation and boundary condition checks in C.

* Enqueue condition: $(\text{rear} + 1) \% \text{MAX} == \text{front}$ (Full).

* De

Q3. Explain how recursion is implemented using the Call Stack with an example (Factorial / Tower of Hanoi).

Step-by-step memory trace diagram showing Activation Records (Stack Frames) pushed during recursive calls and popped during return.